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B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0604 HEAT AND MASS TRANSFER (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand the basic modes of heat transfer.

CO2: Get an insight on conduction, convection and radiation heat transfers.

CO3: Apply the knowledge of heat transfer through various geometries for optimizing the thickness of insulation.

CO4: Understand the unsteady heat conduction and its applications.

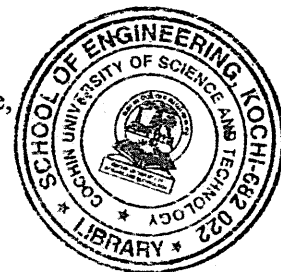
CO5: Get a concept on multiphase flow, diffusion and convective mass transfer.

CO6: Attain information of parallel and counterflow heat exchangers and their design aspects.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome



PART A (Answer *ALL* questions)

I.	(8 × 3 = 24)	Marks	BL	CO	PO
(a) Explain unsteady state heat conduction with an example.	3	3	L2	4	1
(b) Consider two finned surfaces that are identical except that the fins on the first surface are formed by casting whereas they are attached to the second surface afterwards by welding. Identify the case in which the fins will provide greater enhancement in heat transfer. Explain.	3	3	L3	1	3
(c) State the governing law for Diffusion mass transfer.	3	3	L1	5	1
(d) Explain development of hydrodynamic and thermal boundary layers with suitable figure.	3	3	L2	2	1
(e) Define shape factor and state its physical significance.	3	3	L1	2	1
(f) Explain radiation shields and mention some of their applications.	3	3	L2	2	1
(g) Show the temperature variation along the length of heat exchanger when hot and cold fluids flow in parallel and counter flow arrangement.	3	3	L1	6	1
(h) What do you mean by "Fouling" in heat exchangers? How does it affect the performance of heat exchanger?	3	3	L1	6	1

(P.T.O.)

PART B

(4 × 12 = 48)

- | | | | | | | |
|-----------|-----|--|---|----|---|---|
| II. | (a) | Explain the significance of critical radius of insulation. | 4 | L2 | 3 | 1 |
| | (b) | A steam pipe 75 mm inside diameter and 90 mm outside diameter is lagged with two successive layers of insulation. The layer in contact with the pipe is 38 mm asbestos and the asbestos layer is covered with 25 mm thick magnesia insulation. The surface coefficients for inside and outside surfaces are 227 W/m ² K and 6.8 W/m ² K respectively. If the steam temperature is 375°C and the ambient temperature is 35°C, calculate the steady state loss of heat from steam for 60 m length of pipe. Determine the overall coefficient of inside and outside surfaces of the lagged steam pipe. Thermal conductivity values are:
Pipe material - 45 W/mK
Asbestos - 0.14 W/mK
Magnesia Insulation - 0.07 W/mK | 8 | L4 | 3 | 2 |
| OR | | | | | | |
| III. | (a) | Derive equations of temperature distribution and heat dissipation for an infinitely long fin. | 7 | L3 | 3 | 1 |
| | (b) | A motor body is 360 mm in diameter (outside) and 240 mm long. Its surface temperature should not exceed 55°C when dissipating 340 W. Longitudinal fins of 15 mm thickness and 40 mm height are proposed. The convection coefficient is 40 W/m ² °C. Determine the number of fins required. Atmospheric temperature is 30°C, thermal conductivity = 40 W/m°C. | 5 | L4 | 3 | 2 |
| IV. | (a) | Atmospheric air at 25°C flows over a flat plate at a velocity of 3 m/s. If the plate is 1m wide and 80°C, calculate the following at x = 300 mm.
(i) Hydrodynamic boundary layer thickness
(ii) Thermal boundary layer thickness
(iii) Local friction coefficient
(iv) Average friction coefficient
(v) Local heat transfer coefficient
(vi) Average heat transfer coefficient
(vii) Heat transfer. | 7 | L3 | 2 | 2 |
| | (b) | An open pan 20 cm in diameter and 8 cm deep contains water at 25°C and is exposed to dry atmospheric air. If the rate of diffusion of water vapour is 8.54 × 10 ⁴ kg/hr, estimate the diffusion coefficient of water in air. | 5 | L3 | 5 | 2 |
| OR | | | | | | |
| V. | (a) | A condenser is to be designed to condense 225 Kg of steam per hour at a pressure of 0.15 bar. A square array of 400 tubes, each of 6 mm in diameter, is available for the task. If the tube surface temperature is to be maintained at 26°C, make calculations for the length of each tube. | 6 | L3 | 2 | 2 |
| | (b) | Explain diffusivity, diffusional resistance and mass transfer coefficient. Write the units for each term. | 6 | L3 | 5 | 1 |

(Continued)

BTS-VI(R)-04.22-0303

VI.	(a)	Two concentric cylinders have inner and outer radius with 5 cm and 10 cm and length 20 cm. Calculate the view factors.	6	L3	2	2
	(b)	Determine the heat lost by radiation per meter length of 80 mm diameter pipe at 300°C, if	6	L4	2	2
		(i) located in a large room with red brick walls at temperature of 27°C				
		(ii) enclosed in a 160 mm diameter red brick conduit at a temperature of 27°C.				
		Emissivities of pipe material and brick 0.79 and 0.93 respectively.				
		OR				
VII.	(a)	Explain the terms absorptivity, reflectivity and transmissivity.	3	L2	2	1
	(b)	Two large plates, one at 800 K and other at 600 K have emissivities 0.5 and 0.8 respectively. A radiation shield having an emissivity 0.1 on one side and emissivity 0.05 on the other side is placed between the plates. Obtain the heat transfer by radiation per square meter with and without the radiation shield.	9	L4	2	2
VIII.	(a)	Derive an equation for the effectiveness (ϵ) of counter flow heat exchanger in terms of NTU and Capacity Ratio (C).	8	L3	6	1
	(b)	A feed water heater uses steam condensing at 120°C to heat water from 30°C to 90°C in a shell and tube heat exchanger at a rate of 5 kg/s. The overall heat transfer coefficient is equal to 2000 W/m ² K. Due to usage deposits have formed on the surfaces resulting in a fouling resistance of 0.0009 Km ² /W. Determine the water exit temperature if flow rates and inlet temperatures are maintained.	4	L3	6	2
		OR				
IX.	(a)	Derive the expression for LMTD of a parallel flow heat exchanger.	4	L3	6	1
	(b)	A counter flow heat exchanger is used to cool 2000 kg/hr of oil ($C_p = 2.5$ kJ/kgK) from 105°C to 30°C by the use of water entering at 15°C. The overall heat transfer coefficient is expected to be 1.5 KW/m ² K. Presume that the exit temperature of the water is not to exceed 80°C. Using NTU method, make calculations for the	8	L3	6	2
		(i) water flow rate				
		(ii) surface area required for heat transfer				
		(iii) effectiveness of heat exchanger.				

Bloom's Taxonomy Levels

L1 – 16.6 %, L2 – 20.8 %, L3 – 45.8 %, L4 – 16.6 %, L5 – 0 % and L6 – 0 %.

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B.Tech. Degree VI Semester Regular/Supplementary Examination June 2023

ME 19-205-0604 HEAT AND MASS TRANSFER (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the basic modes of heat transfer.

CO2: Get an insight on conduction, convection and radiation heat transfers.

CO3: Apply the knowledge of heat transfer through various geometries for optimizing the thickness of insulation.

CO4: Understand the unsteady heat conduction and its applications.

CO5: Get a concept on multiphase flow, diffusion and convective mass transfer.

CO6: Attain information of parallel and counterflow heat exchangers and their design aspects.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	State and explain Fourier's law of heat conduction.	3	L2	1	1
	(b)	Define Biot number. What is its significance?	3	L1	4	1
	(c)	How we can differentiate thermal and hydrodynamic boundary layers?	3	L2	5	1
	(d)	Why are higher heat transfer rates experienced in drop wise condensation than in film condensation?	3	L2	5	1
	(e)	State and explain Kirchhoff's law.	3	L1	2	1
	(f)	Explain the terms radiosity and irradiation with a simple sketch.	3	L1	2	1
	(g)	What do you mean by fouling factor? Mention the causes of fouling.	3	L2	6	1
	(h)	Compare the parallel flow and counterflow heat exchanger. Why the counterflow heat exchangers are commonly used?	3	L2	6	1

PART B

(4 × 12 = 48)

II. (a) Derive two-dimensional heat conduction equation in Cartesian coordinates. State the assumptions used. Represent the same in vector form.	6	L3	1,2	2
(b) A composite wall of a house is made of 0.1 m layer of common brick ($k = 0.7 \text{ W/m}^\circ\text{C}$) followed by a 0.04 m layer of gypsum plaster ($k = 0.48 \text{ W/m}^\circ\text{C}$). What thickness of loosely packed rock wool insulation ($k = 0.065 \text{ W/m}^\circ\text{C}$) should be added to reduce the heat loss or gain through the wall by 80%?	6	L3	1,2	2

OR

III. (a) State and explain the critical thickness of insulation based on electrical wires and a steam pipe. Derive an expression for critical thickness of insulation for a cylinder.	6	L3	3	2
(b) A wire of 7 mm diameter at a temperature of 65°C is to be insulated by a material having thermal conductivity $k = 0.174 \text{ W/m}^\circ\text{C}$. The convective heat transfer coefficient $h_o = 8.712 \text{ W/m}^2 \text{ }^\circ\text{C}$. The ambient temperature is 20°C . Find the critical radius of insulation. For maximum heat loss, what is the minimum thickness of insulation and the heat loss per metre length? Also compare the amount of heat dissipated with this thickness of insulation and without insulation.	6	L3	3	3

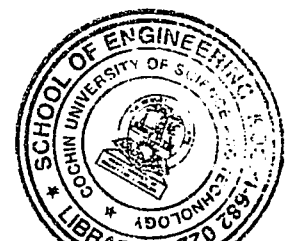
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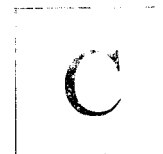
		Marks	BL	CO	PO
IV.	(a) Explain the boiling regimes with a suitable sketch.	6	L2	5	2
	(b) Water at the rate of 0.6 kg/s is passed through a tube of 20 mm diameter. Then it is found to be heated from 20°C to 50°C. The heating is accomplished by condensing steam on the surface of the tube and subsequently the surface temperature of the tube is maintained at 85°C. Determine the length of the tube required for fully developed flow. Take the thermos physical properties of water at 60°C as: $\rho = 983.2 \text{ kg/m}^3$; $C_p = 4.178 \text{ kJ/kgK}$, $k = 0.659 \text{ W/m}^\circ\text{C}$; kinematic viscosity $\nu = 0.478 \times 10^{-6} \text{ m}^2/\text{s}$.	6	L3	5	2
OR					
V.	(a) Define Reynolds Analogy.	5	L1	5	1
	(b) A cylindrical body of 300 mm diameter and 1.6 m height is maintained at a constant temperature of 36.5°C. The surrounding temperature is 13.5°C. Find out the amount of heat to be generated by the body per hour if: $\rho = 1.025 \text{ kg/m}^3$, $C_p = 0.96 \text{ KJ/kg}^\circ\text{C}$, $\nu = 15.06 \times 10^{-6} \text{ m}^2/\text{s}$, $k = 0.0892 \text{ kJ/m-sec}^\circ\text{C}$, and $\beta = 0.003356 \text{ K}^{-1}$. Assume $Nu = 0.12 (\text{Gr. Pr})^{1/3}$	7	L3	5	2
VI.	(a) Write a note on the properties of black body. What potential changes in heat transfer occur if a surface selected is gray instead of black surface?	5	L2	2	1
	(b) Assuming the sun to be a black body having a surface temperature of 5800 K, calculate: (i) the total emissive power (ii) the wavelength at which the maximum spectral intensity occurs (iii) the maximum value of $E_{b\lambda}$ (iv) the total energy emitted by the sun per unit time if its diameter can be assumed to be $1.391 \times 10^9 \text{ m}$.	7	L3	2	2
OR					
VII.	(a) Explain the effect of using radiation shield. What is the criteria for selecting shield materials?	4	L2	2	1
	(b) Two large plates are placed parallel, one at temperature $t_1 = 827^\circ\text{C}$ with emissivity $\epsilon_1 = 0.8$ and other at $t_2 = 327^\circ\text{C}$ with emissivity $\epsilon_2 = 0.4$. An aluminium radiation shield with an emissivity $\epsilon_s = 0.05$ on both sides is placed between the plates. Calculate the percentage reduction in heat transfer rate between the two plates as a result of the shield.	8	L3	2	2
VIII.	(a) Derive the expression for LMTD for a counter flow heat exchanger.	6	L2	6	2
	(b) The flow rates of hot and cold water streams running through a parallel flow heat exchanger are 0.2 kg/s and 0.5 kg/s respectively. The inlet temperatures on the hot and cold sides are 75°C and 20°C respectively. The exit temperature of hot water is 40°C. If the individual heat transfer coefficients on both sides are 650 W/m ² °C, calculate the area of the heat exchanger.	6	L3	6	3
OR					
IX.	(a) How are the heat exchangers classified?	2	L1	6	1
	(b) In an oil cooler oil enters at 170°C. If water entering at 30°C flows parallel to the oil, the exit temperature of oil and water are 90°C and 70°C respectively. Determine the exit temperature of oil and water if the two fluids flow in the opposite directions. Assume that the flow rates of the two fluids and U_o remain unaltered. What would be the minimum temperatures to which the oil could be cooled in parallel flow operation?	10	L3	6	2

Bloom's Taxonomy Levels

L1 = 20.83%, L2 = 37.5%, L3 = 41.67%.



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B.Tech. Degree VI Semester Special Supplementary Examination January 2019

ME 15-1604 HEAT AND MASS TRANSFER (2015 Scheme)

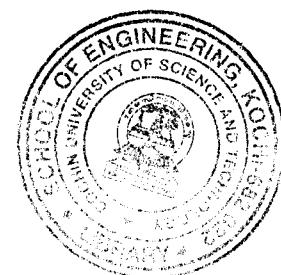
Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain Fourier law of heat conduction and the concept of thermal resistance.
- (b) What is critical thickness of insulation? Explain its significance.
- (c) Explain significance of fin effectiveness and fin efficiency.
- (d) Explain thermal boundary layer for forced convection heat transfer from flat plates.
- (e) Discuss drop wise and film wise condensations.
- (f) Explain radiation shape factor and shape factor algebra.
- (g) Explain electromagnetic radiation spectrum.
- (h) Explain Planck's law and Wien's displacement law.
- (i) How heat exchangers are classified based on the flow direction?
- (j) Define heat exchanger effectiveness and NTU.



PART B

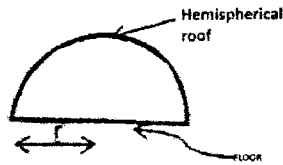
(4 × 10 = 40)

- II. (a) Derive general heat conduction equation in cylindrical coordinates. (6)
 - (b) Calculate the critical radius of insulation for asbestos [$k = 0.17 \text{ W/m}^\circ\text{C}$] surrounding a pipe and exposed to room air at 20°C with $h = 3.0 \text{ W/m}^2\text{C}$. Also determine the heat loss per unit length from a 200°C , 5 cm diameter pipe, when covered with the critical radius of insulation and without insulation. (4)
- OR**
- III. (a) Derive the expression for temperature distribution and heat transfer rate in an infinitely straight fin of rectangular profile. (6)
 - (b) A steel ball [$C_p = 0.46 \text{ kJ/kg}^\circ\text{C}$, $k = 35 \text{ W/m}^\circ\text{C}$, $\rho = 7800 \text{ kg/m}^3$], 5 cm in diameter and initially at a uniform temperature of 450°C is suddenly placed in a controlled environment in which the temperature is maintained at 100°C . The convection heat-transfer coefficient is $10 \text{ W/m}^2\text{C}$. Calculate the time required for the ball to attain a temperature of 150°C . (4)
- IV. (a) Air at 27°C and 1 atm flows over a flat plate at a speed of 2 m/s. The plate is heated over its entire length to a temperature of 60°C . (i) Calculate the hydrodynamic and thermal boundary-layer thickness at distances of 20 cm and 40 cm from the leading edge of the plate. (ii) Calculate the heat transferred from top surface in the first 20 cm of the plate. (iii) Calculate the skin friction or drag coefficient at a distance of 40 cm from leading edge of plate. (6)
 - (b) What is Reynold's analogy? Describe relation between fluid friction and heat transfer. (4)

OR

(P.T.O.)

- V. (a) A horizontal pipe 0.3 m in diameter is maintained at a temperature of 250°C in a room where the ambient air is at 15°C . Calculate the free-convection heat loss per meter of length. (5)
- (b) Explain the boiling regimes and boiling curve with the help of a suitable sketch. (5)
- VI. (a) Derive the relation for radiative heat exchange between two non-black infinite parallel planes. (6)
- (b) For a hemispherical furnace, the floor is at 700 K, and has an emissivity of 0.5. The hemispherical top is at 1000 K and has emissivity of 0.25. Determine the net radiative heat transfer per unit area from roof to floor. (4)



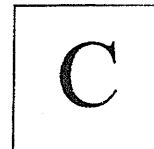
OR

- VII. (a) Calculate the net radiant heat exchange per m^2 area for two large parallel plates at temperatures of 727°C and 127°C respectively. The emissivities of hot and cold plates are 0.8 and 0.4 respectively. If a polished aluminium shield of emissivity 0.04 on both sides is placed between them, find the shield temperature and percentage reduction in the heat transfer. (7)
- (b) Explain the terms absorptivity, reflectivity and transmissivity. (3)
- VIII. (a) A counter flow heat exchanger is used to cool 2000 kg/hr of oil ($C_p = 2.5 \text{ kJ/kgK}$) from 105°C to 30°C by the use of water entering at 15°C . The overall heat transfer coefficient is expected to be $1.5 \text{ kW/m}^2\text{K}$. Presume that the exit temperature of the water is not to exceed 80°C . Using NTU method, make calculations for: (i) the water flow rate (ii) the surface area required for heat transfer (iii) the effectiveness of heat exchanger. (7)
- (b) What do you mean by fouling factor? What are the causes of fouling? (3)

OR

- IX. (a) Derive an expression for log mean temperature difference (LMTD) for a parallel flow heat exchanger. (5)
- (b) Steam condenses at 100°C on the external surface of the tubes of a steam condenser. The tubes are 12 in number and each is 30 mm in diameter and 10 m long. The inlet and outlet temperatures of cooling water flowing inside the tubes are 25°C and 60°C . If the water flow rate is 1.1 kg/s, calculate (i) Rate of condensation of steam (ii) The overall heat transfer coefficient based on the tube surface area. (5)

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B.Tech. Degree VI Semester Special Supplementary Examination November 2022

ME 19-205-0604 HEAT AND MASS TRANSFER

(2019 Scheme)

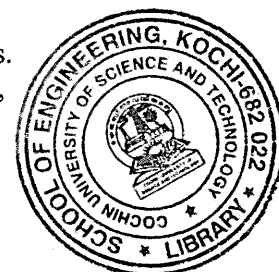
Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the basic modes of heat transfer.
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 CO3: Apply the knowledge of heat transfer through various geometries for optimizing the thickness of insulation.
 CO4: Understand the unsteady heat conduction and its applications.
 CO5: Get a concept on multiphase flow, diffusion and convective mass transfer.
 CO6: Attain information of parallel and counter flow heat exchangers and their design aspects.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome



PART A (Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	Write short notes on critical thickness of Insulation.	3	L3	3	1
(b)	Show that the temperature profile for the heat conduction through a wall of constant thermal conductivity is a straight line.	3	L2	2	1,2
(c)	Write short notes on Ficks law of diffusion.	3	L1	5	1,2,3
(d)	Compare film wise and drop wise condensation.	3	L2	1	1,2,3
(e)	State and prove Kirchhoff's identity.	3	L2	4	1,2,3
(f)	Write short notes on Radiation shields.	3	L2	4	1,2,3
(g)	Explain NTU method of heat exchanger analysis.	3	L3	6	1,2,3
(h)	Write short notes on Classification of Heat exchanger.	3	L1	6	1,2,3

PART B

(4 × 12 = 48)

- II. A hollow cylinder with inner radius 30 mm and outer radius 50 mm is heated at the inner surface at a rate of 105 W/m^2 and dissipated heat by convection from outer surface into a fluid at 80°C with heat transfer coefficient of $400 \text{ W/m}^2\cdot\text{K}$. There is no energy generation and thermal conductivity of the material is constant at $15 \text{ W/m}\cdot\text{K}$. Calculate the temperatures of inside and outside surfaces of the cylinder.
- OR**
- III. A person is found dead at 5 p.m. in a room where temperature is 20°C . The temperature of the body is measured to be 25°C when found, and the heat transfer coefficient is estimated to be $8 \text{ W/m}^2\cdot\text{K}$. Modelling the human body a 30 cm diameter, 1.70 m long cylinder, calculate actual time of death of the person. Take thermo physical properties of the body:
- $k = 6.08 \text{ W/m}\cdot\text{K}$, $\rho = 900 \text{ kg/m}^3$, $C = 4000 \text{ J/kg}\cdot\text{K}$.

(P.T.O.)

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		Marks	BL	CO	PO
IV.	Air at 10°C and at a pressure of 100 kPa is flowing over a plate at a velocity of 3 m/s. If the plate is 30 cm wide and at a temperature of 60°C. Calculate the following quantities at $x = 0.3$ m. (i) Boundary layer thickness (ii) Local friction coefficient (iii) Local shearing stress (iv) Total drag force (v) Thermal boundary layer thickness (vi) Local convective heat transfer coefficient (vii) The heat transfer from the plate, if the length of the plate is 0.5 m.	12	L4	5	1,2,3
OR					
V.	Consider a rectangular plate 0.2 m × 0.4 m is maintained at a uniform temperature of 80°C. It is placed in atmospheric air at 24°C. Compare the heat transfer rates from the plate for the cases when the vertical height is (i) 0.2 m (ii) 0.4 m.	12	L4	5	1,2,3
VI.	A plane, gray, diffuse and opaque surface (absorptivity = 0.7) with a surface area of 0.5 m ² , is maintained at 500°C and receives radiant energy at a rate of 10,000 W/m ² . Determine per unit time: (i) The energy absorbed (ii) The radiant energy emitted (iii) The total energy leaving the surface per unit area (iv) The radiant energy emitted by the surface in the wave band 0.2 μm to 4 μm (v) The net radiative heat transfer from the surface.	12	L3	2	1,2,3
OR					
VII.	Two concentric cylinders have inner and outer radii 5 cm and 10 cm, respectively and length 20 cm. Calculate the all possible view factors.	12	L3	4	1,2,3
VIII.	A heat exchanger is required to cool 55,000 kg/h of alcohol from 66°C to 40°C using 40,000 kg/h of water entering at 5°C. Calculate: (i) exit temperature of water (ii) heat transfer rate (iii) surface area required for: (1) parallel flow type (2) counter flow type of heat exchanger. Take overall heat transfer coefficient $U = 580$ W/m ² .K, C_p (alcohol) = 3760 J/kg.K C_p (water) = 4180 J/kg.K.	12	L5	6	1,2,3
OR					
IX.	A chemical having specific heat of 3.3 kJ/kg.K at a rate of 20,000 kg/h enters a parallel flow heat exchanger at 120°C. The flow rate of cooling water is 50,000 kg/h with an inlet temperature of 20°C. The heat transfer area is 10 m ² and overall heat transfer coefficient is 1050 W/m ² .K. Find: (i) The effectiveness of the heat exchanger (ii) Outlet temperature of water and chemical. Take C_p of water as 4.186 kJ/kg.K	12	L5	6	1,2,3

Bloom's Taxonomy Levels

L1 = 8.33%, L2 = 16.6%, L3 = 58.3%, L4 = 33.33%, L5 = 50%.

B.Tech. Degree VI Semester Examination April 2018

ME 15-1604 HEAT AND MASS TRANSFER (2015 Scheme)

(Approved heat and mass transfer data books are permitted)

Time: 3 Hours

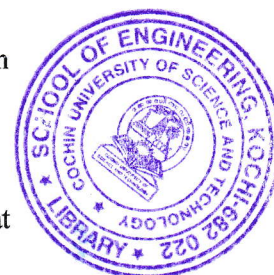
Maximum Marks: 60

PART A

(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) Explain thermal conductivity. Derive the thermal conduction resistance for a composite plane wall with three different materials in series.
- (b) Derive an expression to find the critical radius of insulation over a cylinder. Give an example where you prefer to have critical radius for the insulation.
- (c) What is the significance of biot number in the lumped system analysis of unsteady conduction? Explain the limit of biot number applicable for a lumped system.
- (d) Draw the boundary layers for a hot vertical plate suspended in atmospheric air. Mark velocity and temperature profiles.
- (e) Explain different stages of pool boiling with pool boiling graph.
- (f) What is meant by a gray body? What is its significance in radiation analysis?
- (g) Explain emissivity. What is the emissivity of sun?
- (h) Explain the significance of NTU for heat exchangers.
- (i) Draw the temperature profile variation for parallel flow/counter flow heat exchangers and condenser.
- (j) Explain the effect of fouling on heat exchanger performance. How does it affect the design of a heat exchanger?



PART B

(4 × 10 = 40)

- II. (a) Consider a cylindrical 0.10 cm diameter copper thermocouple wire originally at 150°C. Determine the time required for the thermocouple to reach equilibrium with 1°C difference with the fluid when this wire is suddenly immersed in water at 40°C (heat transfer coefficient 80 W/m²K) and in air at 40°C (heat transfer coefficient 10 W/m²K). (5)
 - (b) Compare the heat loss from an insulated and an uninsulated copper pipe under the following conditions. The pipe (conductivity 400 W/mK) has an internal diameter of 10 cm and an external diameter of 12 cm. Saturated steam flows inside the pipe at 110°C. The pipe is located in a space at 30°C and the heat transfer coefficient on its outer surface is estimated to be 15 W/m²K. The insulation available to reduce heat losses is 5 cm thick and its conductivity is 0.20 W/mK. (5)
- OR**
- III. (a) A steel pipe line ($k = 50$ W/mK) of ID 100 mm and OD 110 mm is to be covered with two layers of insulation each having a thickness of 50 mm. The thermal conductivity of the first insulation material is 0.06 W/mK and that of the second is 0.12 W/mK. Calculate the loss of heat per metre length of pipe and the interface temperature between the two layers of insulation when the temperature of the inside tube surface is 250°C and that of the outside surface of the insulation is 50°C. (5)

(P.T.O.)

- (b) To increase the heat dissipation from a 2.5 cm OD tube, circumferential fins made of aluminium (conductivity 200 W/mK) are soldered to the outer surface. The fins are 0.1 cm thick and have an outer diameter of 5.5 cm. The tube temperature is 100°C, the environmental temperature is 25°C, and the heat transfer coefficient between the fins and the environment is 65 W/m²K. There are 100 fins per meter length of tube. Calculate the rate of heat loss with fins and without fins (assume same heat transfer coefficient) per meter length of the tube. (5)
- IV. (a) An electronic device is cooled by water flowing through capillary holes drilled in the casing. The temperature of the device casing is constant at 353 K. The capillary holes are 0.3 m long and 2.54×10^{-3} m in diameter. If water enters at a temperature of 333 K and flows at a velocity of 0.2 m/s, calculate the outlet temperature of water. (5)
- (b) Determine the Nusselt number for water flowing at an average velocity of 3 m/s in an annulus formed between a 2.54 cm OD tube and a 34.8 cm ID tube. The water is at 83°C and is being cooled. The temperature of the inner wall is 38°C, and the outer wall of the annulus is insulated. Neglect entrance effects and choose appropriate correlation. (5)
- OR**
- V. (a) Water entering at 10°C is to be heated to 40°C in a tube of 0.02 m ID at a mass flow rate of 0.01 kg/s. The outside of the tube is wrapped with an insulated electric-heating element that produces a uniform flux of 15,000 W/m² over the surface. Neglecting any entrance effects, determine the heat transfer coefficient, the length of pipe needed for a 30°C increase in average water temperature. (5)
- (b) A 2.0 cm diameter horizontal heater is maintained at a surface temperature of 38°C and submerged in water at 27°C. Calculate the free-convection heat loss per unit length of the heater. (5)
- VI. (a) Liquefied oxygen (boiling temperature, -183°C) is to be stored in a spherical container with 0.3 m in diameter. The system is insulated by an evacuated space between the inner sphere and a surrounding 0.46 m ID concentric sphere. Both spheres are made of polished aluminum (emissivity = 0.03), and the temperature of the outer sphere is -1°C. Estimate the rate of heat flow by radiation to the oxygen in the container. (5)
- (b) Two parallel plates 0.5 m by 1.0 m are spaced 0.5 m apart. One plate is maintained at 1000°C and the other at 500 °C. The emissivities of the plates are 0.2 and 0.5, respectively. The plates are located in a very large room, the walls of which are maintained at 27°C. The plates exchange heat with each other and with the room, but only the plate surfaces facing each other are to be considered in the analysis. Find the net transfer to each plate and to the room. (5)
- OR**
- VII. (a) Two very large parallel planes with emissivities 0.3 and 0.8 exchange heat. Find the percentage reduction in heat transfer when a polished-aluminum radiation shield (emissivity 0.04) is placed between them. (5)
- (b) Two large parallel planes having emissivities of 0.3 and 0.5 are maintained at temperatures of 900 K and 400 K, respectively. A radiation shield having an emissivity of 0.05 on both sides is placed between the two planes. Calculate the heat-transfer rate per unit area if the shield were not present, the heat-transfer rate per unit area with the shield present, and the temperature of the shield. (5)

(Contd...3)

- VIII. (a) Determine the heat transfer surface area required for a heat exchanger constructed from a 0.0254 m OD tube to cool 6.93 kg/s of an ethyl alcohol solution (Specific heat $C_p = 3810 \text{ J/kg K}$) from 65.6°C to 39.4°C , using 6.306 kg/s of water available at 10°C ($C_p = 4187 \text{ J/kgK}$). Assume that the overall coefficient of heat transfer based on the outer-tube area is $568 \text{ W/m}^2\text{K}$ and consider each of the following arrangements: (i) parallel-flow (ii) counter-flow. Assume that the alcohol is flowing through the shell and the water is flowing through the tubes. (5)
- (b) Hot oil at 100°C is used to heat air in a shell-and-tube heat exchanger. The oil makes six tube passes and the air makes one shell pass; 2.0 kg/s of air are to be heated from 20°C to 80°C . The specific heat of the oil is $2100 \text{ J/kg}^\circ\text{C}$, and its flow rate is 3.0 kg/s. Calculate the area required for the heat exchanger for $U = 200 \text{ W/m}^2^\circ\text{C}$. (5)

OR

- IX. (a) A shell-and-tube heat exchanger is used as an ammonia condenser with ammonia vapor entering the shell at 50°C as a saturated vapor. Water enters the single-pass tube arrangement at 20°C and the total heat transfer rate required is 200 kW. The overall heat-transfer coefficient is estimated as $1000 \text{ W/m}^2^\circ\text{C}$. Determine the area to achieve a heat exchanger effectiveness of 60 percent with an exit water temperature of 40°C . What percent reduction in heat transfer would result if the water flow is reduced to half while keeping the heat exchanger area and U the same? (6)
- (b) Derive an expression to find the LMTD of a countreflow heat exchanger. (4)

